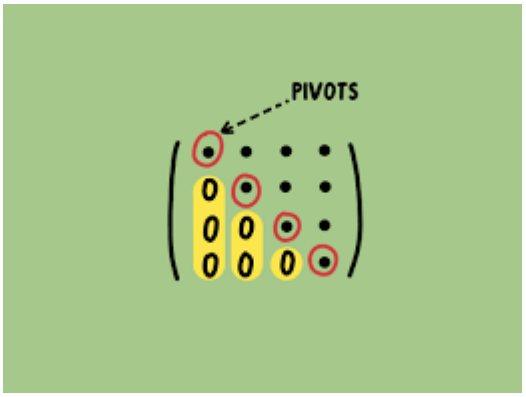
Row reduction



Systems of equations to matrices

Original system

$5a + b = 17$

$4a - 3b = 6$

→

Intermediate System

$a + 0.2b = 3.4$

$b = 2$

→

Solved system

$1a + 0b = 3$

$0a + 1b = 2$

Original matrix

5	1
4	-3

→

Upper diagonal matrix

1	0.2
0	1

Row echelon form

→

Diagonal matrix

1	0
0	1

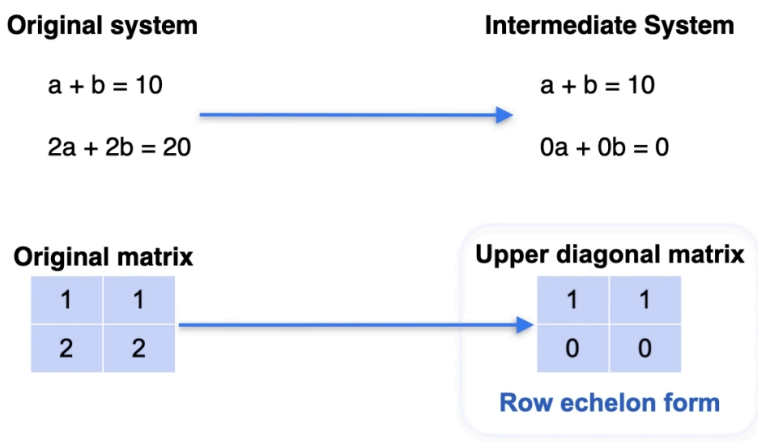
Reduced row echelon form

Upper Triangular Matrix

$$U = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ 0 & a_{22} & a_{23} & a_{24} \\ 0 & 0 & a_{33} & a_{34} \\ 0 & 0 & 0 & a_{44} \end{bmatrix}_{4\times4}$$

Lower Triangular Matrix

$$L = \begin{bmatrix} a_{11} & 0 & 0 & 0 \\ a_{21} & a_{22} & 0 & 0 \\ a_{31} & a_{32} & a_{33} & 0 \\ a_{41} & a_{42} & a_{43} & a_{44} \end{bmatrix}_{4\times4}$$

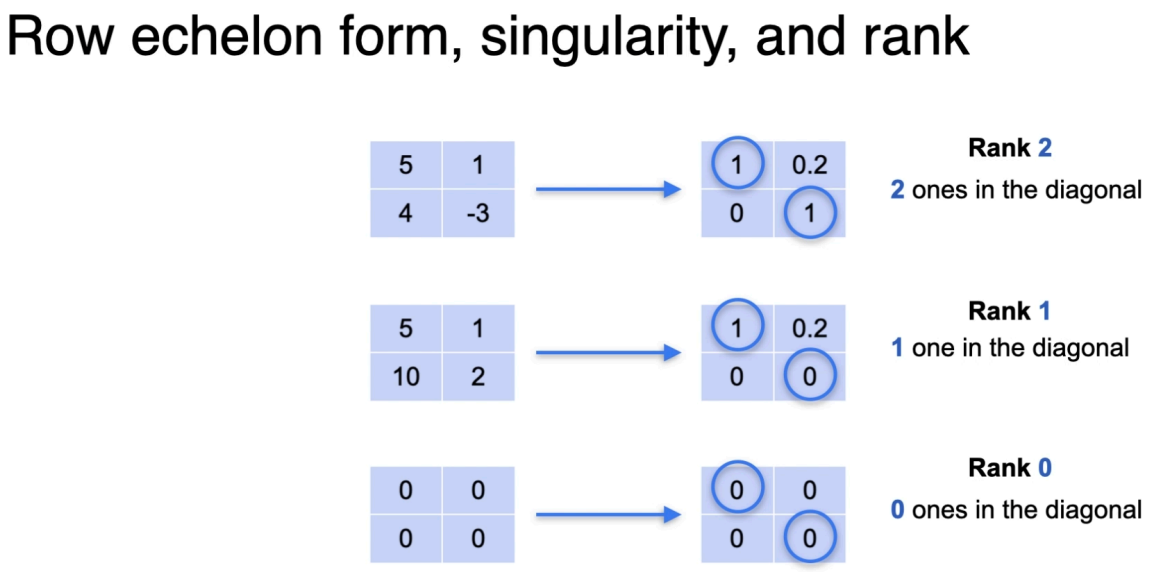


Row Echelon form

1	*	*	*	*
0	**1**	*	*	*
0	0	**0**	0	0
0	0	0	**0**	0
0	0	0	0	**0**

1	*
0	**1**

1	*
0	**0**

0	0
0	**0**

The pivot can be all 1, all 0, or 1 followed by a bunch of 0s.